

Bi-Level Inter-Modality Modulation for Unsupervised Visible-Infrared Person Re-Identification

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Abstract—The task of unsupervised visible-infrared person re-identification (USL-VI-ReID) aims to retrieve cross-modal pedestrian images without manual annotations. The key challenge lies in achieving semantic alignment to resolve modality bias in the absence of real labels. However, existing methods overly rely on single-modal information in the process of pseudo-label generation without considering cross-modal associations, making it difficult to bridge the modality gap between visible and infrared images. To address these issues, this paper proposes a Bi-level Inter-Modal Modulation Network (BIMM-Net), which employs multi-level cluster structure optimization as a core strategy to drive the establishment of cross-modal semantic associations, ultimately achieving cross-modal alignment at the feature representation level. Specifically, we construct a novel intermediary modality GrayMix from visible images to enhance model robustness against color variations and alleviate modality gaps. To filter out noise in cross-modal matching and establish a shared semantic space between visible and infrared modalities, we further develop a Ternary Pairs Calibration-Convergence module designed for filtering noise from visible-infrared cluster matching, on this basis constructing fused mixture clusters. Building on this mixture cluster space, an Heterogeneous-Isomorphic Alignment Loss is also designed to align the feature distributions of the three modalities, reinforcing cross-modal semantic consistency. In addition, we present a Cross-modal Neighborhood Consistency Clustering method, which facilitates the formation and propagation of cross-modal clusters by selecting high-confidence cross-modal neighbor pairs and refining feature distances. Ultimately, BIMM-Net through the joint modeling of bi-level clustering enables multiple levels to guide each other in refining cross-modal structures, thereby effectively establishing the semantic associations between visible and infrared modalities.

Extensive experiments validate the superior performance of the proposed framework, achieving state-of-the-art results in USL-VI-ReID. The source code of this paper is available at: <https://github.com/liujunyu5920/DIMM-Net>

Index Terms—Person re-identification, cross-modal retrieval, semantic alignment, clustering.

I. INTRODUCTION

VISIBLE-INFRA-RED Person Re-Identification (VI-ReID) focuses on cross-modal retrieval of pedestrian images, where the objective is to retrieve the corresponding pedestrian images in the infrared (IR) gallery given a query image from the visible (RGB) modality, or vice versa. Compared to traditional single-modal Re-ID tasks that rely solely on visible cameras, VI-ReID faces a fundamental challenge due to the significant modality gap between visible and infrared images. In the VI-ReID task, several excellent works [1], [2], [3] have achieved promising performance on labeled VI-ReID datasets, but their strong reliance on large-scale manual annotations severely limits their scalability and real-world applicability. Consequently, unsupervised visible-infrared person re-identification (USL-VI-ReID) [4] has emerged as a crucial research direction that demands further exploration.

USL-VI-ReID does not rely on manual labels but instead uses clustering algorithms (e.g., DBSCAN [5], K-means [6]) to automatically generate pseudo-labels. Some studies [7], [8], [9], [10] conduct separate clustering for visible and infrared samples and construct memory banks by storing cluster centers, then optimize feature representations through memory drift strategies and contrastive learning to promote intra-cluster compactness and inter-cluster separability. However, the lack of cross-modal interaction leads to semantic heterogeneity between modalities for samples of the same identity, which limits performance. To address this issue, some methods [7], [11], [12] further matched visible and infrared sample pairs, as establishing cross-modal correspondences is critical for bridging the feature distribution gap between modalities. Although extensive efforts are devoted to building reliable cross-modal matching pairs, the matching accuracy remains constrained by the lack of subsequent filtering and refinement steps, relying solely on basic algorithms such as the Hungarian [13] algorithm or optimal transport [14]. Nevertheless, the substantial modality gap continues to constrain the performance of these approaches. One alternative strategy [15] jointly clusters

Received 15 August 2025; revised 2 January 2026 and 4 February 2026; accepted 20 February 2026. Date of publication 5 March 2026; date of current version 13 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62501226 and Grant 62576067, in part by the Natural Science Foundation of Hebei Province under Grant F2025201037, in part by the Basic Research Project of Shijiazhuang Municipal Universities in Hebei Province under Grant 241791387A, in part by the National Key Research and Development Program of China under Grant 2024YFB4710800, in part by Liaoning Provincial Natural Science Foundation under Grant 2025-YQ-01 and Grant 2024-MS-012, and in part by Dalian Science and Technology Talent Innovation Support Plan under Grant 2024RY010. The associate editor coordinating the review of this article and approving it for publication was Dr. Xiangbo Shu. (Jinjia Peng and Junyu Liu contributed equally to this work.) (Corresponding author: Huibing Wang.)

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Digital Object Identifier 10.1109/TIFS.2026.3671087

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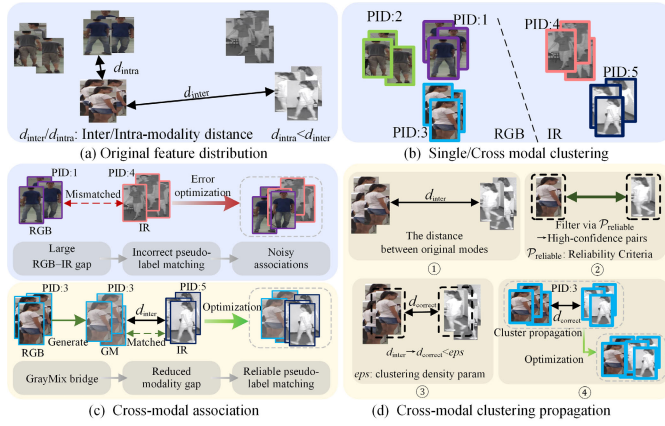


Fig. 1. (a) and (b) illustrate the feature distributions and clustering results of existing unsupervised VI-ReID methods. (c) compares direct cross-modal association with the proposed strategy that performs association via GrayMix as an intermediary. (d) demonstrates how the proposed method overcomes the limitation of cross-modal clustering in forming truly cross-modal clusters.

samples from both modalities to form cross-modal clusters and optimizes the combined memory bank. This strategy partially mitigates modality discrepancy, but some mixture clusters still contain samples from only one modality. Furthermore, some methods [16], [17], [18] propose pseudo-label correction strategies that attempt to more accurately estimate the probability of each sample being noise by mining the neighbor information of samples, thereby improving the reliability of pseudo-labels and providing more effective supervision signals for subsequent feature learning and cross-modal matching.

To overcome these issues, we propose a Bi-level Inter-Modal Modulation Network (BIMM-Net). The motivation of BIMM-Net is illustrated in Fig. 1. As shown in Fig. 1(a), due to the inherent modality gap, RGB and IR samples form separated feature distributions, where inter-modality distances are significantly larger than intra-modality ones, making direct cross-modal matching unreliable. As a result, as illustrated in Fig. 1(b), existing single-modal or direct cross-modal clustering strategies tend to produce modality-dominated clusters and fail to form true cross-modal groups containing semantically consistent RGB and IR samples.

Fig. 1(c) further shows how such large inter-modal distances introduce noise into cross-modal association and pseudo-label generation when RGB and IR samples are directly matched. To mitigate this issue, we introduce GrayMix as an intermediate modality that reduces effective inter-modal distance while preserving structural information, enabling more reliable cross-modal associations. Building on GrayMix, a Ternary Pairs Calibration-Convergence module refines RGB-IR and GrayMix-IR cluster relations and fuses consistent matches into a mixture memory. Finally, as illustrated in Fig. 1(d), Cross-modal Neighborhood Consistency Clustering propagates high-confidence associations across modalities by correcting feature distances, allowing cluster membership to spread across RGB and IR and facilitating the formation of genuinely cross-modal clusters. Through multi-level optimization over association, alignment, and clustering, BIMM-Net achieves robust noise suppression and stable cross-modal semantic consistency. The main contributions of the paper are summarized as follows:

- To address modality bias, we propose a Bi-level Inter-Modal Modulation Network based on multi-level modeling, which leverages intra-modal and inter-modal information to generate high-quality pseudo labels, improving cross-modal semantic alignment.
- The proposed Ternary Pairs Calibration-Convergence module leverages GrayMix as an intermediary modality to mediate cross-modal discrepancies, enhancing the robustness of cross-modal matching and improving semantic alignment by filtering noise and refining cluster fusion.
- To enforce the consistency of samples from different modes in the feature space, we define a Heterogeneous-Isomorphic Alignment Loss that embeds their probability distributions into a shared subspace to achieve multi-modal alignment.
- This paper proposes a Cross-modal Neighborhood Consistency Clustering method optimized with Mutual Nearest Neighbor conditions to refine sample relationships and facilitate the generation of cross-modal clusters.

II. RELATED WORK

A. Visible-Infrared Person Re-Identification

Due to the fundamental differences in imaging principles between RGB and infrared cameras, one of the primary challenges in supervised visible-infrared person re-identification lies in bridging the modality gap. Existing studies typically address this discrepancy either at the image level or the feature level. At the image level, various augmentation strategies [19] have been proposed to reduce modality differences. For instance, CAJ [20] proposed a random channel exchange augmentation which randomly selects one channel from an RGB image and copies its values to the other two channels, producing an image that contains only single-channel information. This augmentation effectively removes color information and aids in mitigating the modality gap with infrared images. DMA [21] explored a similar idea by randomly swapping any two channels of an RGB image to simulate color variations, thereby improving the model's robustness to color perturbations. Another image-level strategy employs adversarial learning (e.g., GAN) to generate pseudo cross-modal samples for mining modality-invariant information. D²RL [1] mitigated modal discrepancies by converting VIS images and IR images into each other. JSIA-ReID [2] proposed a strategy to generate paired cross-modal images and align them at both global and instance levels. However, such approaches incur additional computational overhead during sample generation and struggle to ensure the discriminability of the synthesized images.

In contrast, feature-level approaches aim to embed features from both modalities into a shared feature space through explicit constraints. MMD-ReID [22] proposed an Maximum Mean Discrepancy [23] loss tailored for the VI-ReID task, which reduces modality discrepancy by minimizing the similarity distance between the two modalities in the Reproducing Kernel Hilbert Space at the identity level. GMS [24] introduced the use of the Earth Mover's Distance (EMD) [25] to measure modality differences, minimizing the EMD between RGB and IR features to further reduce distribution gaps. Although these methods have achieved impressive performance on fully annotated datasets, their heavy reliance on

high-quality annotations limits the practical deployment and scalability of VI-ReID in real-world applications.

B. Unsupervised Visible-Infrared Person Re-Identification

USL-VI-ReID has emerged to address this challenge, eliminating the need for labor-intensive manual annotations and instead relying entirely on the network to autonomously determine sample identity labels. The key bottleneck in the development of USL-VI-ReID lies in how to generate robust pseudo-labels and establish reliable cross-modality correspondences. The H2H [4] method used a homogeneous-to-heterogeneous learning framework for USL-VI-ReID. The CHCR [26] method tackled fully USL-VI-ReID via cross-modality hierarchical clustering and pseudo-label refinement without intra-modality initialization. PGM [7] proposed the Progressive Graph Matching algorithm, which constructs a cross-modality cost matrix and applies the Hungarian algorithm for cluster matching, ensuring that each unimodal cluster is paired with a corresponding cluster from the other modality. Additionally, a cross-modality contrastive learning scheme [27] was introduced to bridge the modality gap. However, PGM overlooks the impact of noise, which limits its performance. Building upon PGM, MMM [15] introduced multi-cluster memory for each identity, aiming to simultaneously construct intra-modality clusters and cross-modality clusters to enable finer-grained matching. However, when forming cross-modality clusters, MMM relies on DBSCAN with a fixed density threshold, which is difficult to accommodate the large distribution discrepancy between visible and infrared modalities. As a result, the obtained clusters are often dominated by a single modality, making it challenging to form reliable cross-modal clusters that genuinely contain samples from both modalities, and thus leading to limited robustness of multiple pseudo labels for the same identity under modal bias. The SDCL [17] framework explores shallow-deep collaborative learning for unsupervised visible-infrared person re-identification by jointly leveraging shallow and deep feature representations. Through collaborative neighbor learning and ranking association, SDCL exploits cross-layer consistency to facilitate cross-modality alignment beyond relying solely on deep features. The DOTLA [16] framework addressed unsupervised VI-ReID by using dual optimal transport for cross-modality label assignment and neighbor consistency for label refinement.

Existing USL-VI-ReID methods primarily rely on clustering algorithms to generate pseudo-labels, with strategies including separate modality clustering, cross-modal matching via basic algorithms, joint clustering, and pseudo-label correction. However, these approaches still lack high-quality modeling of cross-modal semantic associations and rely heavily on intra-modality information for pseudo-label generation, making it difficult to alleviate the semantic gap between visible and infrared modalities. Unlike existing approaches, the Bi-level Inter-Modal Modulation Network (BIMM-Net) proposed in this paper innovatively conducts bi-level modulation across modalities. In the single-modal clustering level, BIMM-Net corrects the noise in cross-modal label establishment and embeds features from different modalities into a unified feature subspace for alignment, thereby eliminating modality discrepancies. In the cross-modal clustering level, BIMM-Net

promotes the generation and propagation of clusters between modalities. These two aspects synergistically complement each other, enabling effective cross-modal alignment.

III. THE BI-LEVEL INTER-MODAL MODULATION NETWORK

A. Overview

To explicitly model cross-modal semantic associations and mitigate matching noise as well as suboptimal cluster generation in the multi-level clustering framework, this paper proposes the Bi-level Inter-Modal Modulation Network (BIMM-Net), as illustrated in Fig. 2. The overall model takes the first stage training result of the baseline [7] as the initial model, following the baseline settings. DBSCAN [5] is independently applied to the visible and infrared modalities to generate modality-specific memories, and the model is optimized using contrastive learning [28] and memory updating strategies to obtain initial discriminative features.

Building on the baseline, BIMM-Net (Fig. 2) proposes four core components to bridge RGB and IR domains in a unified multi-level clustering framework. First, the intermediary GrayMix modality is constructed by linearly interpolating the Channel-Aggregated Grayscale Base into each RGB channel, which narrows the color gap with the infrared modality while preserving structural details. Next, the Ternary Pairs Calibration-Convergence (Fig. 2 (a)) module refines and merges RGB-IR and GrayMix-IR matching pairs into mixture memory. A Heterogeneous-Isomorphic Alignment loss (Fig. 2 (b)) then aligns RGB, IR, and GrayMix features to their mixture memory. Finally, Cross-modal Neighborhood Consistency Clustering (Fig. 2 (c)) selects high-confidence neighbor pairs, sets their distances to zero, and drives cross-modality cluster propagation—forming a synergistic loop that progressively mitigates modality bias.

B. GrayMix Intermediary Modality

Given an unlabeled visible-infrared dataset, the visible samples are denoted as $X^v = \{x_v^i\}_{i=1}^{N_v}$ and the infrared samples as $X^r = \{x_r^j\}_{j=1}^{N_r}$, where N_v and N_r represent the total number of visible and infrared images, respectively. Here, each x_v^i denotes the i -th visible image and each x_r^j denotes the j -th infrared image. Each image corresponds to a specific pedestrian identity, but the identity labels are unavailable during the training phase. The objective of USL-VI-ReID is to design an effective feature learning and cross-modality matching strategy that enables the model to retrieve cross-modality images of the same identity as the query image during the testing phase. Due to the significant modality gap between visible and infrared images, directly performing cross-modality matching is extremely challenging. Therefore, it is essential to develop robust optimization strategies and training pipelines to handle modality discrepancy, unlabeled scenarios, and clustering noise.

Building on this, we first design a GrayMix (GM) to serve as a bridge for alleviating the modality gap between visible and infrared light. Fundamentally, the “color” in an RGB image arises from the differences and combinations among its three channel intensities: each pixel is defined by its red, green, and blue components, and a network learns to

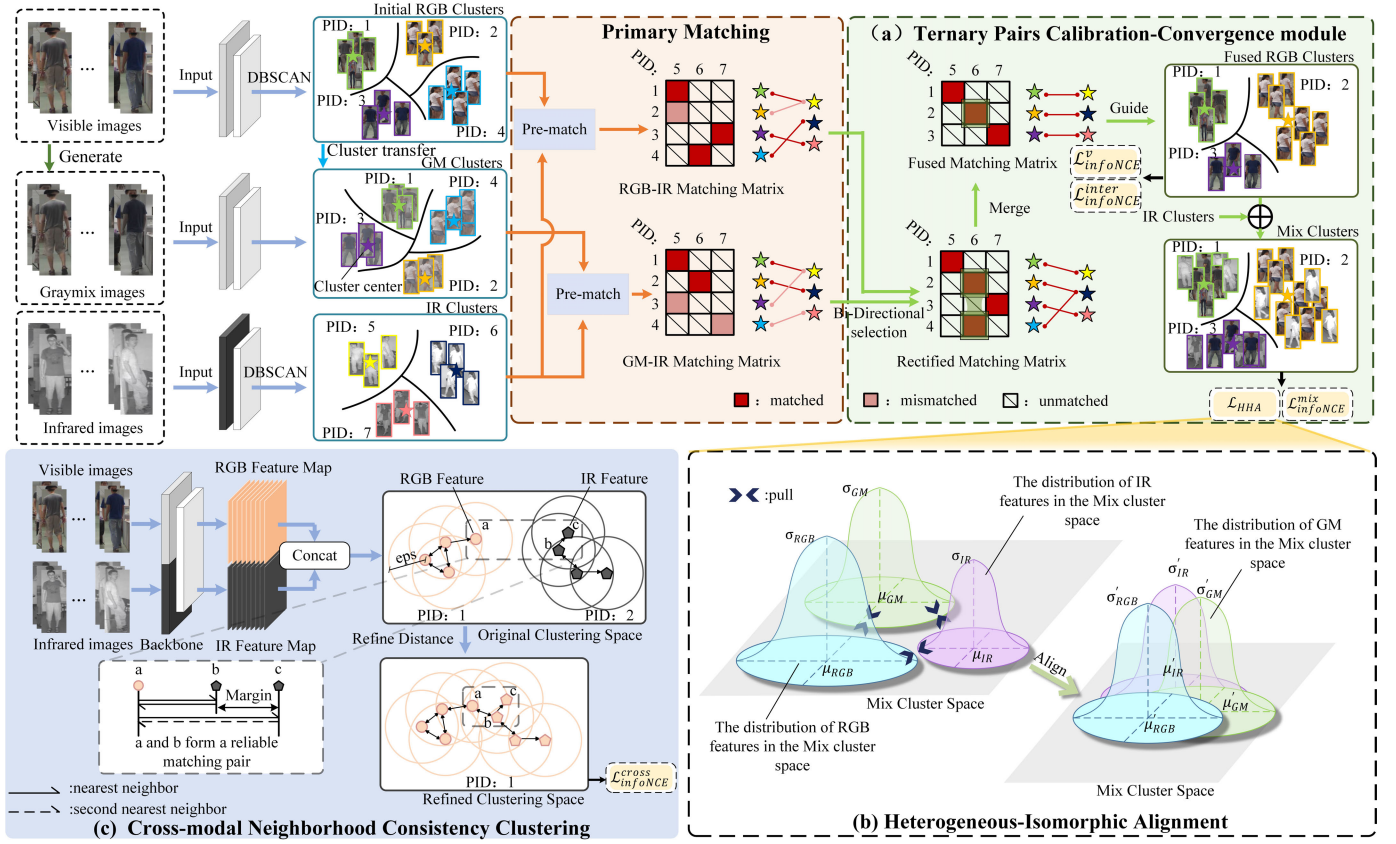


Fig. 2. The overview of the proposed BIMM-Net, which contains a two-level structure. At the single-modal clustering level, it includes a novel intermediary modality image GrayMix, a Ternary Pairs Calibration-Convergence module, and a Heterogeneous-Isomorphic Alignment Loss. At the cross-modal clustering level, it incorporates a novel Cross-modal Neighborhood Consistency Clustering method.

distinguish colors by modeling these inter-channel numerical variations. In contrast, an infrared image contains only a single channel and is typically replicated across all three channels for compatibility with convolutional architectures, resulting in identical values on each channel and therefore lacking the inter-channel variation that encodes color. This structural difference in channel construction underlies the core modality gap between visible-spectrum and infrared images.

Existing methods employ the Channel Augmented (CA) [29] strategy to generate an auxiliary modality. Fundamentally, CA randomly selects a single channel from the RGB image and replaces the other two channels with it, forcing the CA modality samples to be projected into the same single-channel subspace as the IR images at the input stage. CA completely removes the influence of color information, encouraging the model to focus on learning modality-invariant features, which effectively alleviates the training difficulty in the early phase of unsupervised learning. However, the absence of color information and multi-channel data limits the potential for further performance improvement.

To address this, we propose the GrayMix intermediary modality. Unlike CA, GrayMix first obtains the Channel-Aggregated Grayscale Base by fusing information from each channel, and then linearly interpolates this Channel-Aggregated Grayscale Base into three channels to reduce the differences between channels. While weakening color information, GrayMix effectively alleviates modality discrepancies.

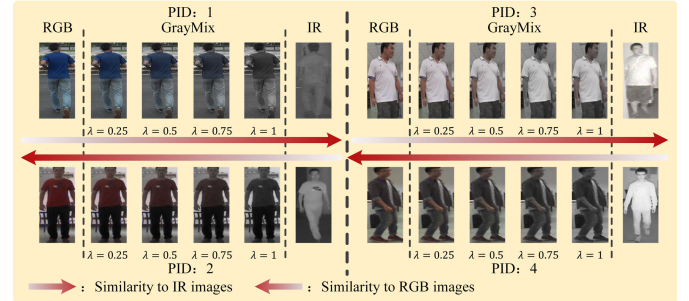


Fig. 3. Visualization of GrayMix images under different λ values. When λ is small, it has a higher similarity to RGB images. However, when λ is large, its color information is missing, and it shows higher similarity to IR images.

The effect of GrayMix is shown in Fig. 3. The intensity of color information can be determined by adjusting the linear interpolation parameters, thereby achieving a balance between weakening and retaining color information.

On the original three-channel RGB image, GrayMix can be formally interpreted as a linear transformation in the color space. For each pixel, we denote the RGB vector as:

$$\mathbf{x} = \begin{bmatrix} R \\ G \\ B \end{bmatrix} \in \mathbb{R}^3 \quad (1)$$

where R, G, and B represent the pixel values of the three channels, and $\mathbb{R}^3$ represents the continuous three-dimensional

color space in which each dimension corresponds to one color channel. The Channel-Aggregated Grayscale Base can be expressed as:

$$\text{Gray} = \frac{1}{3} \mathbf{1}^\top \mathbf{x}, \quad \mathbf{1} = [1, 1, 1]^\top \quad (2)$$

which corresponds to projecting $\mathbf{x}$ onto the luminance subspace spanned by $\mathbf{1}$. Essentially, *Gray* is a compressed form of the color information—it preserves the total brightness but discards the proportion among the three color components. This projection can be written in matrix form as:

$$\mathbf{P} = \frac{1}{3} \mathbf{1} \mathbf{1}^\top \quad (3)$$

where $\mathbf{P}$ is a symmetric and idempotent matrix, indicating that it defines an orthogonal projection onto a one-dimensional color-invariant subspace.

GrayMix then constructs the transformed pixel as:

$$\text{GrayMix} = (1 - \lambda)\mathbf{x} + \lambda\mathbf{P}\mathbf{x} = ((1 - \lambda)\mathbf{I} + \lambda\mathbf{P})\mathbf{x} \quad (4)$$

The parameter λ acts as the trade-off regulator via its dual control over the two components in Eq. 4. The $(1 - \lambda)\mathbf{x}$ term preserves the original inter-channel differences and color-dependent discriminative features, which are critical for maintaining pedestrian discrimination capability. The $\lambda\mathbf{P}\mathbf{x}$ term removes channel-wise variations and enforces consistency across RGB channels, approximating the single-channel characteristic of infrared data and reducing the structural discrepancy between the two modalities. Varying λ continuously within $[0, 1]$ modulates the balance between the two objectives. A small λ emphasizes color and detail preservation to maintain feature discriminability, and a large λ strengthens cross-modal homogeneity to alleviate the modality gap.

As shown in Fig. 3, as λ increases, the interpolated three-channel values progressively converge toward a unified grayscale value. This leads to reduced image saturation, diminished color diversity, and effectively suppresses color differences, which helps to alleviate the modality gap between visible and infrared images. Moreover, since λ is randomly sampled, GrayMix can theoretically produce an unlimited number of intermediate images with varying levels of color saturation from the same input sample, which is critical for enhancing the model's robustness to color variations. Importantly, because the interpolation retains a proportion of the original R, G, and B values, key structural information such as brightness gradients, textures, and edges remains well preserved. Unlike direct grayscale conversion, this process maintains the global contrast and richness of the multi-channel data.

C. Ternary Pairs Calibration-Convergence

Cross-modal contrastive learning is crucial for narrowing the modality gap and boosting cross-modal recognition performance, with effective cluster matching as its core prerequisite. However, due to inherent differences between visible and infrared modalities, existing methods [7], [15], [30], [31] generate substantial matching noise when directly matching their clusters, severely hindering cross-modal learning. To address this, we propose the Ternary Pairs Calibration-Convergence (TPCC) module. GrayMix lies between visible and infrared

modalities in terms of appearance and representation, enabling it to serve as an effective intermediary for mitigating cross-modal discrepancies. Leveraging GrayMix as an intermediary modality that bridges visible and infrared modalities, TPCC first obtains RGB-IR matching pairs, then further matches GrayMix-IR. Using this intermediary property, the module filters and corrects noise in both sets of matching pairs, fuses the final results to construct mixture clusters, and thus embeds cross-modal samples into the same semantic space.

Prior to implementing the TPCC module, this paper clusters the visible and infrared modalities respectively using the DBSCAN [5] algorithm to obtain their pseudo-labels:

$$Y_i^t = \text{DBSCAN}(f(x_i^t)) \quad (5)$$

where Y denotes the pseudo labels generated by clustering, $t \in \{v, r\}$ indicates the modality (visible or infrared), and $f(\cdot)$ represents the backbone. To maintain the relative stability of pseudo labels, at the beginning of each epoch, the memory for each cluster is initialized by computing the mean of all features within the cluster, as follows:

$$c_y^t = \frac{1}{N_y} \sum_{i=1}^{N_y} (f(x_i^t)) \quad (6)$$

where c_y^t denotes the memory corresponding to label y , and N_y represents the number of samples within the cluster labeled as y .

1) *Primary Matching Stage*: TPCC first models the cross-modal memory matching problem as a weighted bipartite graph matching task:

$$\begin{aligned} & \min_{\mathbf{M}} C^T \mathbf{M} \\ & \text{s.t. } \forall i \in [1, N^A], \forall j \in [1, N^B] : m_{ij} \in \{0, 1\}, \\ & \forall i \in [1, N^A] : \sum_{j=1}^{N^B} m_{ij} \leq 1, \\ & \forall j \in [1, N^B] : \sum_{i=1}^{N^A} m_{ij} = 1. \end{aligned} \quad (7)$$

where $\mathcal{A}$ and $\mathcal{B}$ refer to the two modalities in the RGB-IR setting, each taking the form of either the visible modality or the infrared modality. $\mathbf{M} = \{m_{ij}\} \in \{0, 1\}^{N^A \times N^B}$ is a binary indicator matrix, with each element m_{ij} denoting the matching relationship between the i -th node c_i^A in the source modality $\mathcal{A}$ and the j -th node c_j^B in the target modality $\mathcal{B}$. Specifically:

$$m_{ij} = \begin{cases} 1, & \text{if } c_i^A \text{ and } c_j^B \text{ correspond to the same individual,} \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

Meanwhile, in Eq. 7, C represents the cost matrix, which is composed of the reciprocals of the exponential values of the cosine similarities of cross-modal pairs:

$$C_{ij} = \frac{1}{\exp(\cosine(c_i^A, c_j^B))}, \quad \forall i \in [\mathcal{A}], \forall j \in [\mathcal{B}] \quad (9)$$

where $\exp(\cdot)$ denotes the exponential function and $\cosine(\cdot, \cdot)$ computes the cosine similarity between the cluster representations c_i^A and c_j^B from modalities $\mathcal{A}$ and $\mathcal{B}$, respectively.

To solve the optimal matching problem associated with the above-mentioned cost matrix, the Hungarian algorithm [13] is employed as a classical solution. Due to the potential mismatch in cluster numbers between source and target modalities, the above process iteratively performs multiple rounds of Hungarian matching until all clusters find their corresponding cross-modal pairs.

Although the above approach achieves preliminary cross-modal matching through iterative Hungarian algorithm, inherent modality gaps still limit matching accuracy. Given that GrayMix effectively bridges feature distribution disparities across modalities, its integration into the matching pipeline to enhance cross-modal alignment accuracy presents significant research value.

2) *Calibration Stage*: Different from traditional pair matching, TPCC optimizes cross-modal matching results and filters out noise by introducing GrayMix as an intermediate modulation feature. Specifically, it propagates the clustering labels from the RGB modality to GrayMix samples accurately, while constructing memory for each GrayMix cluster:

$$Y_i^{gm} = Y_i^v \quad (10)$$

$$c_y^{gm} = \frac{1}{N_y^{gm}} \sum_{i=1}^{N_y^{gm}} (f(x_i^{gm})) \quad (11)$$

where Y_i^{gm} denotes the assigned pseudo-label of the i -th GrayMix sample, Y_i^v is the pseudo-label of the corresponding visible image, c_y^{gm} represents the memory feature of the GrayMix cluster y , and N_y^{gm} is the number of GrayMix samples belonging to cluster y . x_i^{gm} is the i -th GrayMix image. Based on the rules defined in Eq. 7, the Hungarian algorithm is employed to perform matching operations on RGB-IR and GrayMix-IR cross-modal pairs respectively. For the RGB-IR cross-modal matching pairs, they are denoted as:

$$\mathcal{M}_{RGB-IR} = \{(y_i^v, y_j^r) \mid i \in [1, N^v], j \in [1, N^r]\} \quad (12)$$

where y_i^v represents the clustering label of the i -th memory in the RGB modality, and y_j^r denotes the clustering label of the j -th memory in the infrared modality, and N^v and N^r represent the number of memories in the RGB and infrared modalities, respectively. For the GrayMix/IR cross-modal matching pairs, the representation is:

$$\mathcal{M}_{GM-IR} = \{(y_i^g, y_j^r) \mid i \in [1, N^g], j \in [1, N^r]\} \quad (13)$$

where y_i^g represents the clustering label of the i -th memory in the GrayMix modality, and N^g is the number of memories in the GrayMix modality. It is worth noting that, as shown in Fig. 2, due to the inherent differences in feature space embedding between GrayMix samples and RGB samples, the corresponding cost matrix structure is not completely consistent with that of the RGB modality. This in turn leads to potential inconsistencies in the IR clusters matched by RGB clusters and GrayMix clusters with the same pseudo-labels. To this end, TPCC filters the matching pairs according to the following rules to obtain final results with higher confidence:

$$\mathcal{M}_{rectify} = \mathcal{M}_v \cup \mathcal{M}_g \quad (14)$$

$$\begin{aligned} \mathcal{M}_v &= \{(y_i^v, y_j^r), C_{ij} \leq C_{ik} \vee j = k\} \\ \mathcal{M}_g &= \{(y_i^g, y_k^r), C_{ij} > C_{ik}\} \end{aligned} \quad (15)$$

where $\mathcal{M}_{rectify}$ denotes the rectified cross-modal matching pairs. Specifically, $\mathcal{M}_v$ consists of RGB-IR matching pairs inherited from $\mathcal{M}_{RGB-IR}$ that satisfy the cost constraint $C_{ij} \leq C_{ik}$ or the index consistency condition $j = k$, while $\mathcal{M}_g$ is formed by selecting GrayMix-IR matching pairs from $\mathcal{M}_{GM-IR}$ when $C_{ij} > C_{ik}$. The indices i , j , and k span the label spaces of the RGB modality ($[1, N^v]$) and IR modality ($[1, N^r]$), respectively. Taking cost as the measurement criterion, the matching pairs with lower costs are selected as the final results by comparing the matching costs between RGB/GrayMix modality clusters with the same pseudo-labels and their corresponding IR clusters. The cost function essentially reflects the error degree of cross-modal cluster matching. A lower cost function value indicates higher cross-modal matching consistency, thus enabling effective filtering of noisy matching pairs.

3) *Convergence Stage*: Additionally, due to the frequent discrepancy in the number of clusters between RGB/GrayMix and IR modalities, the “many-to-one” scenario inevitably occurs where multiple source modality clusters match to the same target modality cluster. When multiple source modality clusters match to a single target modality cluster, the semantic information from different source clusters conflicts, leading to ambiguous semantic definitions of the target cluster. Meanwhile, the repeated encoding of the target cluster by these source clusters introduces redundant representations in the feature space, affecting the accuracy and robustness of cross-modal matching. Therefore, when multiple source modality clusters $c_{i_1}^v, c_{i_2}^v, \dots, c_{i_m}^v$ match to the same target modality c_j^r cluster, TPCC defines the following convergence mechanism:

$$c_i^{v'} = \frac{\sum_{k=1}^m N_{i_k}^v \cdot c_{i_k}^v}{\sum_{k=1}^m N_{i_k}^v}, \quad y_i^{v'} = y_{i_1}^v \quad (16)$$

where $N_{i_k}^v$ denotes the number of visible samples belonging to the k -th source cluster $c_{i_k}^v$, and $y_{i_1}^v$ represents the pseudo-label originally assigned to the first source cluster among $c_{i_1}^v, c_{i_2}^v, \dots, c_{i_m}^v$ that are matched to the same target cluster. Moreover, $c_i^{v'}$ denotes the visible memory with label i after convergence, and $y_i^{v'}$ represents the pseudo-label corresponding to the i -th cluster after convergence. Finally, the matching relationship between RGB and IR is:

$$\mathcal{M}'_{RGB-IR} = \{(y_i^{v'}, y_j^r) \mid i \in [1, N^{v'}], j \in [1, N^r]\} \quad (17)$$

where $N^{v'}$ denotes the number of RGB clusters after convergence. Furthermore, to establish a shared embedding space for visible-infrared modalities, an empirical strategy is to fuse the paired RGB and IR clusters to generate mixture clusters. For each pair $(y_i^{v'}, y_j^r)$ in Eq. 17, generate a mixture cluster by fusing them:

$$c_{y_i^{v'} y_j^r}^{mix} = \frac{N_i^{v'} \cdot c_i^{v'} + N_j^r \cdot c_j^r}{N_i^{v'} + N_j^r}, \quad y_i^{mix} = y_i^{v'} \quad (18)$$

where y_i^{mix} denotes the pseudo-label assigned to the Mixture cluster, which is inherited from the converged visible cluster $y_i^{v'}$. Mixture clusters integrate RGB and IR samples with identical pseudo-labels, and the memories constructed from their feature centers encapsulate modality-invariant features learned by the model. By leveraging the mixture memory space as a cross-modal anchoring benchmark, the mixture

contrastive learning strategy effectively bridges the modality gap between visible and infrared modalities. Specifically:

$$\mathcal{L}_{\text{InfoNCE}}^{\text{mix}} = \begin{cases} -\log \frac{\exp(f(x^{v,g}) \cdot c_+^{\text{mix}}/\tau)}{\sum_{i=0}^{N^{\text{mix}}} \exp(f(x^{v,g}) \cdot c_i^{\text{mix}}/\tau)}, & \text{for even epochs} \\ -\log \frac{\exp(f(x^r) \cdot c_+^{\text{mix}}/\tau)}{\sum_{i=0}^{N^{\text{mix}}} \exp(f(x^r) \cdot c_i^{\text{mix}}/\tau)}, & \text{for odd epochs} \end{cases} \quad (19)$$

where c_+ denotes the memory corresponding to the same pseudo-label as x . τ is the temperature coefficient, which controls the smoothness of the similarity distribution. N represents the number of memory entries. The ACCL [7] strategy is adopted to avoid training instability caused by simultaneous optimization of the two modalities. Meanwhile, the inter-modal contrastive learning for visible-infrared clusters also employs the ACCL strategy for optimization:

$$\mathcal{L}_{\text{InfoNCE}}^{\text{inter}} = \begin{cases} -\log \frac{\exp(f(x^{v,g}) \cdot c_+^r/\tau)}{\sum_{i=0}^{N^r} \exp(f(x^{v,g}) \cdot c_i^r/\tau)}, & \text{for even epochs} \\ -\log \frac{\exp(f(x^r) \cdot c_+^v/\tau)}{\sum_{i=0}^{N^v} \exp(f(x^r) \cdot c_i^v/\tau)}, & \text{for odd epochs} \end{cases} \quad (20)$$

In addition, the intra-modal contrastive learning loss is calculated to enhance the discriminative ability within each individual modality:

$$\mathcal{L}_{\text{InfoNCE}}^t = -\log \frac{\exp(f(x^t) \cdot c_+^t/\tau)}{\sum_{i=0}^{N^t} \exp(f(x^t) \cdot c_i^t/\tau)} \quad (21)$$

where $t \in \{v, r\}$ indicates the modality. The total contrastive learning loss is:

$$\mathcal{L}_{\text{InfoNCE}} = \alpha \cdot (\mathcal{L}_{\text{InfoNCE}}^{\text{mix}} + \beta \cdot \mathcal{L}_{\text{InfoNCE}}^{\text{inter}}) + \mathcal{L}_{\text{InfoNCE}}^v + \mathcal{L}_{\text{InfoNCE}}^r \quad (22)$$

where α and β are hyperparameters that balance the contributions of different loss components. we employ a momentum update strategy to optimize the memory:

$$c_y \leftarrow \eta \cdot c_y + (1 - \eta) \cdot f(x_i) \quad (23)$$

where η denotes the momentum coefficient, and $f(x_i)$ represents the feature embedding of the sample extracted by the backbone network.

D. Heterogeneous-Isomorphic Alignment Loss

While TPCC combined with contrastive learning enables some cross-modal sample association in USL-VI-ReID by constructing mixture clusters, it only regulates relative distances between intra-cluster and inter-cluster samples. It fails to constrain the absolute similarity consistency of different modalities with respect to the same mixture cluster center and thus limits exploration of global distribution consistency. To address this, we propose the Heterogeneous-Isomorphic Alignment (HIA) loss. Its core idea is to anchor on mixture memory, measure the distribution distance between each modality's samples and these anchors, and force visible, GrayMix, and infrared samples to form overlapping distributions in mixture

memory space. This mechanism alleviates modality-induced interference in feature alignment.

Before performing distribution alignment, the cross-modal pseudo-paired data generated by Eq. 17 needs to be loaded in each batch to provide the necessary semantic correlation foundation for distribution alignment. Subsequently, the cosine similarity between the features of each modality and the mixture memories are calculated:

$$s(x_i^t, c_j^{\text{mix}}) = \frac{f(x_i^t) \cdot c_j^{\text{mix}}}{\|f(x_i^t)\| \times \|c_j^{\text{mix}}\|} \quad (24)$$

where $t \in \{v, g, r\}$, $(\cdot)$ denotes the dot product, and $\|\cdot\|$ represents the L2 norm. Mapping the original features into a probability distribution in the memory space:

$$p(x_i^t, c_j^{\text{mix}}) = \frac{\exp(s(x_i^t, c_j^{\text{mix}})/\theta)}{\sum_{k=1}^{N^{\text{mix}}} \exp(s(x_i^t, c_k^{\text{mix}})/\theta)} \quad (25)$$

where the temperature parameter θ controls the smoothness of the distribution. This process is equivalent to projecting features into a semantic space anchored by mixture memory, providing a unified measurement benchmark for cross-modal distribution alignment. Concatenate the probability values between B t-mode features and N^{mix} mixture clusters in the batch processing into a matrix:

$$P^t = \begin{bmatrix} p(x_1^t, c_1^{\text{mix}}) & p(x_1^t, c_2^{\text{mix}}) & \cdots & p(x_1^t, c_{N^{\text{mix}}}^{\text{mix}}) \\ p(x_2^t, c_1^{\text{mix}}) & p(x_2^t, c_2^{\text{mix}}) & \cdots & p(x_2^t, c_{N^{\text{mix}}}^{\text{mix}}) \\ \vdots & \vdots & \ddots & \vdots \\ p(x_B^t, c_1^{\text{mix}}) & p(x_B^t, c_2^{\text{mix}}) & \cdots & p(x_B^t, c_{N^{\text{mix}}}^{\text{mix}}) \end{bmatrix} \quad (26)$$

where B denotes the total number of samples from modality t within a training batch. Further, taking samples from RGB and IR modalities as an example, bidirectional KL divergence [32] is used for bidirectional constraint:

$$\mathcal{L}_{\text{HIA}}^{v-r} = \frac{1}{2} (\text{KL}(P^v|P^r) + \text{KL}(P^r|P^v)) \quad (27)$$

Similarly, compute the $\mathcal{L}_{\text{HIA}}^{v-g}$ and $\mathcal{L}_{\text{HIA}}^{g-r}$ between the RGB-GrayMix and GrayMix-IR modalities, and sum them to obtain the final HIA loss:

$$\mathcal{L}_{\text{HIA}} = \mathcal{L}_{\text{HIA}}^{v-r} + \mathcal{L}_{\text{HIA}}^{v-g} + \mathcal{L}_{\text{HIA}}^{g-r} \quad (28)$$

HIA loss function complements the TPCC presented in Sec. III-C. The former addresses the label noise issue in cross-modal matching, while the latter optimizes the distribution consistency of multi-modal features. Together, they form a more comprehensive cross-modal alignment framework. At the single-modal clustering level, the multi-task loss function is defined as:

$$\mathcal{L}_{\text{single}} = \mathcal{L}_{\text{InfoNCE}} + \mu \cdot \mathcal{L}_{\text{HIA}} \quad (29)$$

where μ is the hyperparameter.

E. Cross-Modal Neighborhood Consistency Clustering

As a critical component of the proposed BIMM-Net, the Cross-modal Neighborhood Consistency Clustering method provides reliable pseudo-labels for BIMM-Net at the cross-modal clustering level. Traditional methods using DBSCAN for cross-modal clustering, such as PCAL [33] and

MMM [15], identify cluster structures based on fixed density thresholds. However, in visible-infrared scenarios, inherent modality discrepancies cause the distances between numerous cross-modal sample pairs to exceed the preset threshold ε , leading to the formation of numerous isolated single-modality clusters and failing to achieve effective cross-modal semantic aggregation. For example, in the feature space of infrared and visible images, cross-modal features of the same pedestrian may exhibit large distances due to modality differences, while intra-modal features of different pedestrians may be closer. This makes it difficult for PCAL and MMM to accurately identify cross-modal cluster structures, resulting in a lack of reliable semantic units for subsequent feature alignment.

we propose a method named Cross-modal Neighborhood Consistency Clustering (CNCC) at the cross-modal clustering level, aiming to address the challenge that density-based clustering algorithms [34] struggle to cluster cross-modal samples of the same identity into the same class. Specifically, CNCC first searches for cross-modal nearest neighbors for samples in each modality, and preliminarily selects potential high-confidence pairs through the screening condition of Mutual Nearest Neighbors (MNN):

$$\text{NN}_1(x_i^v, X^r) = \arg \min_{x_j^r \in X^r} \text{cosine}(x_i^v, x_j^r) \quad (30)$$

where x_i^v denotes the i -th visible feature, X^r all infrared features, and $\text{NN}_1(\cdot, \cdot)$ the nearest neighbor operator under cosine distance. The MNN condition for pair (x_i^v, x_j^r) is:

$$\begin{cases} \text{NN}_1(x_i^v, X^r) = x_j^r \\ \text{NN}_1(x_j^r, X^v) = x_i^v \end{cases} \quad (31)$$

The set of high-confidence cross-modal sample pairs filtered by the MNN condition is:

$$\mathcal{P}_{\text{MNN}} = \{(x_i^v, x_j^r) \mid \text{NN}_1(x_i^v, X^r) = x_j^r \wedge \text{NN}_1(x_j^r, X^v) = x_i^v\} \quad (32)$$

After obtaining preliminary high-confidence sample pairs, a distance difference measurement with second nearest neighbors is adopted to reflect relative superiority:

$$\Delta d = d_2 - d_1 = \text{dist}(x_i^v, \text{NN}_2(x_i^v, X^r)) - \text{dist}(x_i^v, \text{NN}_1(x_i^v, X^r)) \quad (33)$$

where $\text{NN}_2(\cdot, \cdot)$ denotes the second nearest neighbor operator across modalities, and the $\text{dist}(\cdot, \cdot)$ denotes the variant of the Jaccard distance defined in [33]. Meanwhile, a consistency score is introduced as an absolute threshold:

$$\text{consistency} = \frac{1}{1 + d_1} = \frac{1}{1 + \text{dist}(x_i^v, \text{NN}_1(x_i^v, X^r))} \quad (34)$$

This dual constraint effectively eliminates false matches caused by local density fluctuations, providing a more robust semantic foundation for cross-modal clustering. A sample pair is considered a reliable association if and only if both conditions are satisfied:

$$\mathcal{P}_{\text{reliable}} = \left\{ (x_i^v, x_j^r) \in \mathcal{P}_{\text{MNN}} \mid \begin{array}{l} \Delta d > \delta \\ \text{consistency} > \varepsilon \end{array} \right\} \quad (35)$$

where δ denotes the distance difference threshold, which ensures the nearest neighbor distance has superiority over the second nearest neighbor. ε is the consistency score threshold, acting as an absolute confidence criterion to filter out unreliable pairs.

Subsequently, the distances between sample pairs in the reliable set $\mathcal{P}_{\text{reliable}}$ are set to zero to facilitate cross-modal cluster propagation. The corrected distance is defined as:

$$d_{\text{corr}}(x_i^v, x_j^r) = \begin{cases} 0, & \text{if } (x_i^v, x_j^r) \in \mathcal{P}_{\text{reliable}} \\ \text{dist}(x_i^v, x_j^r), & \text{otherwise} \end{cases} \quad (36)$$

This strategy forces the distances of high-confidence cross-modal pairs to zero, enabling DBSCAN to treat them as absolute neighbors. This breaks through the distance barrier caused by modality differences and achieves efficient propagation of cross-modal semantic information.

Based on the corrected distances, the $\mathbf{D}_{\text{corr}}$ is fed into the DBSCAN algorithm for clustering:

$$Y^{\text{cross}} = \text{DBSCAN}(f(X^{v,r}), \mathbf{D}_{\text{corr}}) \quad (37)$$

where $\mathbf{D}_{\text{corr}}$ denotes the corrected cross-modal distance matrix whose elements are defined by $d_{\text{corr}}(x_i^v, x_j^r)$, and $Y^{\text{cross}} = \{Y_1^{\text{cross}}, Y_2^{\text{cross}}, \dots, Y_{N^{\text{cross}}}^{\text{cross}}\}$ denotes the set of cross-modal cluster labels, where each label Y_i^{cross} corresponds to a cluster. N^{cross} represents the total number of cross-modal clusters. Finally, the loss function at the cross-modal clustering level is defined as:

$$\mathcal{L}_{\text{InfoNCE}}^{\text{cross}} = -\log \frac{\exp(f(x) \cdot c_+^{\text{cross}} / \tau)}{\sum_{i=0}^{N^{\text{cross}}} \exp(f(x) \cdot c_i^{\text{cross}} / \tau)} \quad (38)$$

where $f(x)$ represents the feature embedding of sample x extracted by the backbone network, c_+ denotes the memory corresponding to the same pseudo-label as x . τ is the temperature coefficient.

F. Optimization Algorithm

To provide a clear and reproducible description of the multi-stage workflow of BIMM-Net, a formal algorithmic specification is presented to detail the complete training procedure, including data flow across modalities, cluster interactions, and stage-wise memory updates.

Alg. 1 outlines the sequential execution of single-modal clustering, GrayMix generation, TPCC, HIA-based feature alignment, and CNCC-based cross-modal propagation, and explicitly specifies how cluster memories are constructed and updated at each stage.

IV. EXPERIMENTS

A. Datasets and Evaluation Metrics

1) *Datasets*: This paper evaluates the proposed approach on two widely used benchmark datasets: SYSU-MM01 [35] and RegDB [36]. As a significant benchmark in the VI-ReID field, the SYSU-MM01 dataset contains pedestrian images from four visible-light and two infrared cameras, with 30,071 visible and 15,792 infrared images. The training data includes 22,258 visible and 11,909 infrared images captured by indoor/outdoor cameras, and evaluation is performed in All-search and Indoor-search modes. The training set and the testing set are independent of each other without overlap. The RegDB dataset, collected via a visible light and thermal dual-camera system, includes 8,240 images in total, with ten visible and ten near-infrared images per identity. It is divided into training and testing subsets of 206 identities each, supporting

Algorithm 1 Bi-level Inter-modality Modulation Network

Input: Unlabeled visible images $X^v = \{x_i^v\}_{i=1}^{N_v}$, infrared images $X^r = \{x_i^r\}_{i=1}^{N_r}$, backbone encoder $f(\cdot)$, training epochs T

Output: Trained network parameters θ

```

1: Initialize  $f(\cdot)$  and memories  $c^v, c^r$  using PGM
2: for  $t = 1$  to  $T$  do
3:   Extract features  $\mathbf{F}^v \leftarrow f(X^v)$ ,  $\mathbf{F}^r \leftarrow f(X^r)$ 
4:   Generate GrayMix samples  $X^g \leftarrow \text{GrayMix}(X^v, \lambda)$ 
5:   Extract GrayMix features  $\mathbf{F}^g \leftarrow f(X^g)$ 
6:   Obtain pseudo labels  $Y^v \leftarrow \text{DBSCAN}(\mathbf{F}^v)$ ,  $Y^r \leftarrow \text{DBSCAN}(\mathbf{F}^r)$ 
7:   Compute cluster memories  $c^v, c^r$  by Eq. (6)
8:   single-modal clustering level(TPCC)
9:   Propagate pseudo labels from RGB to GrayMix samples by Eq. (10)
10:  Construct cluster memories  $c^{gm}$  by Eq. (11)
11:  Perform Hungarian matching on  $(c^v, c^r)$  and  $(c^{gm}, c^r)$ 
12:  Rectify matching pairs via GrayMix by Eq. (14)–(15)
13:  Merge many-to-one RGB clusters by Eq. (16)
14:  Construct mixture memory  $c^{mix}$  by Eq. (18)
15:  Optimize  $\mathcal{L}_{\text{InfoNCE}}$  by Eq. (19)–(22)
16:  Optimize HIA loss  $\mathcal{L}_{\text{HIL}}$  by Eq. (27)–(28)
17:  Update  $c^v, c^r, c^{mix}$  by Eq. (23)
18:  Cross-modal clustering level (CNCC)
19:  Mine reliable RGB–IR pairs by Eq. (31)–(36)
20:  Correct distances by Eq. (37)
21:  Obtain cross-modal clusters  $Y^{\text{cross}}$  by Eq. (38)
22:  Optimize  $\mathcal{L}_{\text{cross}}$  by Eq. (38)
23:  Update cross-modal memory  $c^{\text{cross}}$  by Eq. (23)
24: end for
25: return  $\theta$ 

```

cross-modal evaluations in visible-to-infrared and infrared-to-visible modes, while ensuring complete independence between training and testing data.

2) *Evaluation Metrics:* Following the evaluation protocols in the VI-ReID field, the Cumulative Matching Characteristic (CMC) curve, mean Average Precision (mAP) and mean Inverse Negative Penalty (mINP) are adopted as performance metrics. Moreover, fair comparisons with state-of-the-art methods are conducted to validate the proposed approach.

B. Baseline

The baseline method we adopt is the Progressive Graph Matching (PGM) framework [7], PGM formulates cross-modality correspondence mining at the cluster level by constructing a cost matrix between visible and infrared clusters and applying the Hungarian algorithm to obtain global matchings. To alleviate cluster-number imbalance across modalities, a progressive multi-stage matching strategy is employed to refine the assignments. Based on the mined correspondences, PGM further incorporates an alternate cross-contrastive learning scheme to encourage modality-invariant feature learning while reducing the impact of noisy matches. In our implementation, for the model architecture, a dual-stream ResNet-50 [37] with Non-local [38] modules is employed as the

backbone, and the first-stage result of PGM is used as the pre-trained model.

C. Implementation Details

The BIMM-Net is implemented in PyTorch and trained on a single RTX-4090 GPU. Each training batch loads samples from 8 identities, with 8 samples per modality (three modalities in total), resulting in $8 \times 8 \times 3 = 192$ samples per batch. The image size for both training and testing is set to $288 \times 144 \times 3$. During the formal training phase, the training is conducted with 50 epochs on the SYSU-MM01 dataset and 30 epochs on the RegDB dataset. The initial learning rate is 3.5×10^{-4} , decaying by a factor of 0.1 every 20 epochs. The parameter α follows the Baseline's setting and is set to 0.25. For GrayMix, the grayscale parameter λ (acting as the upper bound) is set to 0.5 for SYSU-MM01 and 1.0 for RegDB; the actual grayscale coefficient used in experiments is randomly drawn from the range $[0, \lambda]$. The temperature parameter τ in Eqs. 19, 20 and 21 and 38 is set to 0.05, while the parameter θ in Eq. 25 is set to 0.1. Additionally, the parameters δ and ε in Eq. 35 are set to 0.008 and 0.6 for the SYSU-MM01 dataset, and 0.008 and 0.9 for the RegDB dataset, respectively. Other hyperparameter values will be provided in the parameter ablation section.

As summarized in Tab. III, BIMM-Net and the baseline share the same backbone and thus have identical model complexity, with 72.60M parameters and 8.19 GFLOPs. Compared with the baseline, BIMM-Net incurs higher training cost, requiring 20.68 GB GPU memory and 7.9 hours on SYSU-MM01, due to the additional bi-level optimization and cross-modal refinement. However, the inference cost remains unchanged: BIMM-Net achieves 136.57 FPS with only 1.88 GB GPU memory. These results indicate that the proposed method mainly increases training overhead while preserving efficient inference, making it suitable for practical deployment once training is completed.

D. Comparison With the State-of-the-Arts

We compare the proposed BIMM-Net with state-of-the-art methods. As shown in Tab. I, BIMM-Net achieves excellent performance across all metrics, establishing a strong benchmark for USL-VI-ReID.

1) *Comparison With Supervised Methods:* Although supervised methods generally outperform unsupervised ones due to the availability of labeled data, recent advances in unsupervised learning have significantly narrowed this gap. Notably, BIMM-Net outperforms several strong supervised baselines such as AlignGAN [40], JSIA-ReID [2], X-Modal [41], DDAG [42], AGW [43]. On SYSU-MM01, although BIMM-Net is an unsupervised method, its performance is already comparable to that of MMD-ReID [22]. On the RegDB dataset, BIMM-Net achieves Rank-1 that are only slightly lower than the top-performing supervised methods, including MMD-ReID [22], CM-EMD [24], DEEN [44] and DMA [21]. BIMM-Net employs multi-level cluster structure optimization to refine cross-modal matching accuracy and enhance inter-modal cluster distribution consistency, ultimately achieving deep alignment of the global semantic space. This mechanism, which directly mines cross-modal commonalities from data without relying on labels, enables robust semantic association

TABLE I

COMPARISONS BETWEEN THE PROPOSED BIMM-NET AND VARIOUS STATE-OF-THE-ART METHODS ON THE SYSU-MM01 AND REGDB DATASETS. RANK AT R ACCURACY (%), MAP (%) AND MINP (%) ARE REPORTED. THE BOLD FONT AND UNDERLINE FONT INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Category	Methods	Venue	SYSU-MM01										RegDB									
			All Search					Indoor Search					VIS to IR					IR to VIS				
			R-1	R-10	R-20	mAP	mINP	R-1	R-10	R-20	mAP	mINP	R-1	R-10	R-20	mAP	mINP	R-1	R-10	R-20	mAP	mINP
SL-VI-ReID	Zero-Pad [35]	ICCV' 2017	14.80	54.12	71.33	15.95	-	20.58	68.38	85.79	26.92	-	17.75	34.21	44.35	18.90	-	16.63	34.68	44.25	17.82	-
	eBDTR [39]	TIFS' 2019	27.82	67.34	81.34	28.42	-	32.46	77.42	89.62	42.76	-	34.62	58.96	68.72	33.46	-	34.21	58.74	68.64	32.49	-
	AlignGAN [40]	ICCV' 2019	42.4	85.0	93.7	40.7	-	45.9	87.6	94.4	54.3	-	57.9	-	-	53.6	-	56.3	-	-	53.4	-
	JSIA-ReID [2]	AAAI' 2020	38.1	80.7	89.9	36.9	-	43.8	86.2	94.2	52.9	-	48.1	-	-	48.9	-	48.5	-	-	49.3	-
	X-Modal [41]	AAAI' 2020	49.9	89.8	96.0	50.7	-	-	-	-	-	-	62.2	83.1	91.7	60.2	-	-	-	-	-	-
	DDAG [42]	ECCV' 2020	54.8	90.4	95.8	53.0	39.6	61.0	94.1	98.4	68.0	62.6	69.3	86.2	91.5	63.5	-	68.1	85.2	90.3	61.8	-
	AGW [43]	TPAMI' 2021	47.50	84.39	92.14	47.65	35.30	54.17	91.14	95.98	62.97	59.23	70.05	86.21	91.55	66.37	-	70.49	87.21	91.84	65.90	-
	MMD-ReID [22]	arXiv' 2021	66.75	94.16	97.38	62.25	-	71.64	97.75	99.52	75.95	-	95.06	98.67	99.31	88.95	-	93.65	97.55	98.38	87.30	-
	CAJ [29]	ICCV' 2021	69.88	95.71	98.46	66.89	53.61	76.26	97.88	99.49	80.37	76.79	85.03	95.49	97.54	79.14	65.33	84.75	95.33	97.51	77.82	61.56
	CM-EMD [24]	AAAI' 2023	73.39	-	-	68.56	-	80.53	-	-	82.71	-	94.37	-	-	88.23	-	92.77	-	-	86.85	-
	DEEN [44]	CVPR' 2023	74.7	97.6	99.2	71.8	-	80.3	99.0	99.8	83.3	-	91.1	97.8	98.9	85.1	-	89.5	96.8	98.4	83.4	-
	DARD [45]	TIFS' 2024	69.33	94.32	97.52	65.65	-	77.21	98.32	99.18	81.91	-	86.19	-	-	85.39	-	85.53	-	-	85.09	-
	DMA [21]	TIFS' 2024	74.57	95.19	97.97	70.41	56.50	82.85	-	-	85.10	-	93.30	-	-	88.34	-	91.50	-	-	86.80	-
	AGPI ² [46]	TIFS' 2025	72.23	97.04	-	70.58	-	83.45	98.62	-	84.25	-	89.03	98.19	-	83.89	-	87.91	97.15	-	83.04	-
USL-VI-ReID	H2H [4]	TIP' 2021	30.15	65.92	77.32	29.30	-	-	-	-	-	-	23.81	45.31	54.00	18.87	-	-	-	-	-	-
	OTLA [11]	ECCV' 2022	29.98	71.79	83.85	27.13	-	29.80	-	-	38.80	-	32.90	-	-	29.70	32.10	-	-	-	28.60	-
	MBCCM [47]	MM' 2023	53.14	89.61	96.74	48.16	32.41	55.21	91.44	95.83	61.98	57.13	83.79	95.83	97.82	77.87	65.04	82.82	<u>95.73</u>	<u>96.89</u>	76.74	61.73
	PGM [7]	CVPR' 2023	57.27	92.48	97.23	51.78	34.96	56.23	90.19	95.39	62.74	58.13	69.48	-	-	65.41	-	69.85	-	-	65.17	-
	MMM [15]	ECCV' 2024	61.60	-	-	57.90	-	64.40	-	-	70.40	-	<u>89.70</u>	-	-	80.50	-	85.80	-	-	77.0	-
	PCAL [33]	TIFS' 2025	57.94	90.98	96.42	52.85	36.90	60.07	93.25	97.27	66.73	62.09	86.43	94.10	96.25	82.51	72.33	86.21	94.05	96.11	81.23	<u>68.71</u>
	DLM [9]	TPAMI' 2025	62.15	<u>93.37</u>	<u>97.44</u>	<u>58.42</u>	<u>43.70</u>	<u>67.31</u>	95.88	99.20	<u>72.86</u>	<u>68.89</u>	87.55	94.66	<u>96.71</u>	82.83	<u>71.93</u>	86.84	94.35	96.37	81.94	68.96
	RoDE [48]	TIFS' 2025	<u>62.88</u>	-	-	57.91	43.04	64.53	-	-	70.42	66.04	88.77	-	-	78.98	67.99	85.78	-	-	78.43	62.34
	BIMM-Net (ours)	-	63.71	93.76	98.06	59.94	45.13	69.02	<u>94.72</u>	<u>97.88</u>	73.84	69.68	89.71	<u>95.39</u>	<u>96.46</u>	<u>82.61</u>	69.91	86.99	95.63	97.33	<u>81.52</u>	67.45

TABLE II

ABLATION STUDY OF DIFFERENT COMPONENTS IN BIMM-NET ON THE SYSU-MM01 AND REGDB DATASETS. RANK-1 ACCURACY (%), MAP (%) AND MINP (%) ARE REPORTED. THE BOLD FONT AND UNDERLINE FONT INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Methods	GrayMix	TPCC	HIA	CNCC	SYSU-MM01 (All Search)			SYSU-MM01 (Indoor Search)			RegDB (VIS to IR)		
					R-1	mAP	mINP	R-1	mAP	mINP	R-1	mAP	mINP
Baseline					57.27	51.78	34.96	56.40	62.74	58.13	69.48	65.41	52.97
A	✓				57.95	54.04	38.53	62.98	68.83	64.24	78.64	74.02	62.15
B	✓	✓			59.22	55.67	40.78	63.51	69.13	64.86	83.25	79.51	68.46
C		✓	✓		59.62	55.89	40.88	62.70	68.81	64.52	84.03	77.18	63.93
D	✓			✓	59.88	55.30	39.48	61.62	67.38	62.81	84.85	80.11	68.24
E	✓	✓	✓		60.63	56.84	42.17	<u>66.12</u>	71.35	67.04	85.97	80.57	68.77
F		✓	✓	✓	<u>61.93</u>	<u>58.20</u>	<u>43.43</u>	66.02	<u>72.06</u>	<u>68.22</u>	<u>87.32</u>	<u>81.11</u>	<u>69.17</u>
G	✓	✓	✓	✓	63.71	59.94	45.31	69.02	73.84	69.68	89.71	82.61	69.91

TABLE III

COMPARISON OF MODEL COMPLEXITY, COMPUTATIONAL COST, AND PERFORMANCE BETWEEN BIMM-NET AND THE BASELINE ON THE SYSU-MM01 DATASET (ALL SEARCH). UNITS: PARAMS (M), GFLOPS, GPU MEM. (GB), TIME (H), FPS, PERFORMANCE (%). THE BOLD FONT AND UNDERLINE FONT INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Method	Model Complexity		Training Cost		Inference Cost		Performance	
	Params	GFLOPs	GPU Mem.	Time	GPU Mem.	FPS	Rank-1	mAP
Baseline	72.60	8.19	15.12	5.3	1.88	136.57	<u>57.27</u>	<u>51.78</u>
BIMM-Net	72.60	8.19	20.68	7.9	1.88	136.57	63.71	59.94

even in the absence of annotations, thus achieving performance comparable to supervised methods.

2) *Comparison With Unsupervised Methods:* Compared with existing unsupervised methods, BIMM-Net achieves clear improvements across all evaluation metrics. On SYSU-MM01 all search mode, it reaches a Rank-1 of 63.71% and mAP of 59.94%, outperforming the second-best method RoDE [48] (62.88% / 57.91%) and DLM [9] (62.15% / 58.42%). Meanwhile, it has also achieved superior performance in indoor search mode. On the RegDB dataset, the Rank-1 of BIMM-Net reached 89.71%, the mAP reached 82.61%, and

most of the other indicators achieved optimal or suboptimal performance.

The above discussion proves the effectiveness of BIMM-Net. The performance gain of BIMM-Net stems from its dual-level modulation design. At the single-modal clustering level, cross-modal matching accuracy is optimized and multi-modal distribution alignment is enhanced to strengthen feature discriminability. At the cross-modal clustering level, inter-modality cluster propagation is optimized to promote the generation of multi-modal clusters. These two levels synergistically guide the learning process, ensuring cross-modal consistency and driving substantial performance improvements.

E. Ablation Studies

In this section, we conduct ablation studies on the components of BIMM-Net to systematically investigate the contribution of each component to the network performance and verify their effectiveness. The results of the ablation experiments are shown in Tab. II.

1) *Effectiveness of GrayMix:* To verify the effectiveness of GrayMix, we introduced this intermediate modality to replace the channel augmentation (CA) mechanism in the

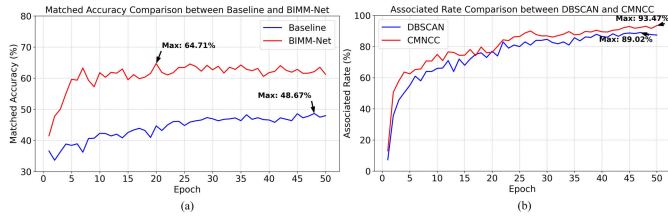


Fig. 4. (a) Comparison of cross-modal matching accuracy between Baseline and BIMM-Net on SYSU-MM01 (All Search). Matched Accuracy is defined as the ratio of matched clusters whose primary labels are identical. (b) Comparison of cross-modal clustering capability on SYSU-MM01 (All Search). Associated Rate denotes the proportion of clusters containing samples from both modalities.

baseline model. As shown by Method A in Tab. II, the model achieves a 0.68% improvement in Rank-1 accuracy, a 2.26% enhancement in mAP, and a 3.57% boost in mINP under the All-Search mode on the SYSU-MM01 dataset. Notable performance gains are also observed in the Indoor-Search mode on the SYSU-MM01 dataset and in the VIS to IR mode on the RegDB dataset. These results demonstrate that GrayMix not only enhances the model's robustness to color variations but also improves cross-domain generalization by serving as an intermediate representation between RGB and IR modalities.

2) *Effectiveness of TPCC*: Tab. II shows that after introducing the TPCC module based on Method A, Method B achieves further improvements in model performance metrics. In the All-Search mode of the SYSU-MM01 dataset, its Rank-1 accuracy reaches 59.22%, while the mAP and mINP metrics increase to 55.67% and 40.78%, respectively. This result indicates that the TPCC module can correct incorrect cross-modal matching pairs as expected, thereby promoting more robust semantic alignment. Additionally, Fig. 4 (a) compares the cross-modal matching accuracy between the Baseline and BIMM-Net, demonstrating that BIMM-Net significantly outperforms the Baseline with a maximum accuracy difference of 16.04%. This further validates the critical role of the TPCC module in handling noisy matching processes.

3) *Effectiveness of HIA*: To further explore the effectiveness of the HIA loss, Method C in Tab. II incorporates the HIA loss on top of TPCC, while Method E adds the HIA loss based on Method B. In the All-Search mode of the SYSU-MM01 dataset, Method C achieves a Rank-1 accuracy of 59.62%, with mAP and mINP reaching 55.89% and 40.88%, respectively. Compared to Method B, Method E demonstrates improvements of 1.41% in Rank-1, 1.17% in mAP, and 1.39% in mINP. The performance gains of both methods collectively validate that the HIA loss effectively aligns the distributions of RGB, IR, and GM modalities in the mixture cluster space, significantly mitigating cross-modal discrepancies. Furthermore, the superior performance of Method E confirms that the effective collaboration among GrayMix, TPCC, and HIL modules can mitigate modality discrepancies and jointly promote cross-modal semantic alignment.

4) *Effectiveness of CNCC*: In Tab. II, Method D (based on Method A) and Method F (based on Method C) incorporate the CNCC module to investigate its contribution to network performance. Experimental results show that Method D achieves a 1.93% improvement in Rank-1 accuracy over Method A in the All-Search mode of SYSU-MM01, with mAP and

TABLE IV
VARIANCE ANALYSIS OF BIMM-NET UNDER DIFFERENT RANDOM SEEDS ON SYSU-MM01. RESULTS ARE REPORTED FOR FIVE INDEPENDENT RUNS. THE BOLD FONT AND UNDERLINE FONT INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Run	All Search			Indoor Search		
	Rank-1	mAP	mINP	Rank-1	mAP	mINP
1	63.82	60.10	45.20	70.12	74.76	70.73
2	63.78	59.40	44.13	68.68	73.88	69.85
3	64.00	59.27	44.08	66.27	71.73	67.45
4	63.24	59.63	45.03	68.72	<u>73.90</u>	<u>69.91</u>
5	63.71	59.94	45.13	69.02	73.84	69.68
Mean $\pm$ Std	63.71 $\pm$ 0.29	59.67 $\pm$ 0.34	44.71 $\pm$ 0.56	68.56 $\pm$ 1.41	73.62 $\pm$ 1.13	69.52 $\pm$ 1.23

mINP increasing by 1.26% and 0.95%, respectively. Method F further boosts Rank-1 to 61.93%, with mAP and mINP reaching 58.20% and 43.43%. The performance gains from CNCC indicate that it effectively promotes cluster propagation across modalities while alleviating modality discrepancy, thus reducing the generation of single-modality clusters caused by modal differences. Furthermore, Fig. 4 (b) compares the cross-modal cluster construction capabilities of DBSCAN and CNCC, demonstrating that CNCC increases the cross-modal cluster generation rate by 4.45%. This further validates that at the cross-modal clustering level, CNCC effectively facilitates the propagation of cluster structures across modalities.

F. Robustness Analysis

1) *Variance Analysis*: To further assess the statistical stability of BIMM-Net, we conduct a multi-seed robustness analysis on the SYSU-MM01 dataset. Specifically, the complete training and evaluation process is repeated several times using different random seeds, and the results are reported for both the All Search and Indoor Search settings. The detailed results, including the mean and standard deviation, are summarized in Tab. IV.

As shown in Tab. IV, BIMM-Net exhibits consistent performance across different random initializations. In the All Search setting, the standard deviation is 0.29% for Rank-1 accuracy and 0.34% for mAP, while in the Indoor Search setting, the standard deviation remains around 1% for both Rank-1 and mAP. The slightly larger variance observed in the Indoor Search setting is mainly due to its smaller scale, where stochastic factors such as clustering initialization and mini-batch sampling have a more pronounced effect.

Overall, the low variance across multiple runs indicates that the performance gains achieved by BIMM-Net are stable and reproducible, rather than being caused by favorable random initialization or clustering randomness. This analysis provides additional support for the reliability of the proposed method on saturated VI-ReID benchmarks.

2) *Domain Shifts*: We further investigate the behavior of BIMM-Net under cross-dataset domain shifts by conducting experiments where the model is trained on SYSU-MM01 and evaluated on RegDB, and vice versa, as reported in Tab. V. Due to the absence of overlapping identities and the significant discrepancies in data distributions, both the baseline and BIMM-Net exhibit low performance in these cross-dataset settings. This observation highlights that severe domain shifts remain a challenging open problem when no explicit domain adaptation or shared identity information is available.

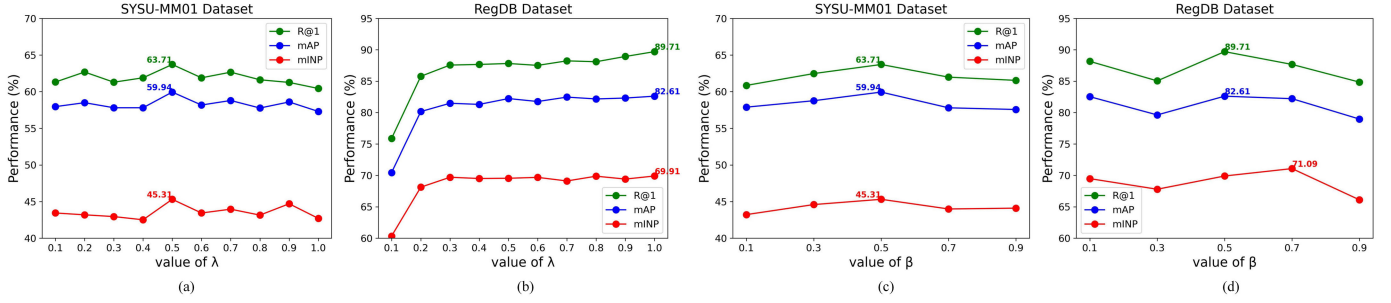


Fig. 5. Influence of λ and β values on SYSU-MM01 dataset under the All Search mode and RegDB dataset under the Visible to Infrared mode. Rank-1 accuracy(%), mAP (%) and mINP (%) are reported.

TABLE V

CROSS-DATASET EVALUATION BETWEEN SYSU-MM01(SYSU) AND REGDB. MODELS ARE TRAINED ON THE SOURCE DATASET AND TESTED ON THE TARGET DATASET. RANK- k ACCURACY (%), MAP (%) AND MINP (%) ARE REPORTED

Method	Task	Rank-1	Rank-5	Rank-10	Rank-20	mAP	mINP
Baseline: PGM [7]	SYSU→RegDB	6.30	11.46	15.00	21.70	7.32	4.75
	RegDB→SYSU	3.91	13.84	21.58	37.49	5.29	2.14
ours: BIMM-Net	SYSU→RegDB	6.46	13.20	17.38	24.08	7.26	4.35
	RegDB→SYSU	4.03	14.04	23.24	37.83	5.66	2.30

TABLE VI

ROBUSTNESS ANALYSIS UNDER UNPAIRED SETTINGS ON SYSU-MM01. CROSS-MODAL ASSOCIATION NOISE IS INJECTED BY RANDOMLY PERTURBING A GIVEN PROPORTION OF CROSS-MODAL PAIRS CONSTRUCTED BY TPCC. THE BOLD FONT AND UNDERLINE FONT INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Noise Ratio	SYSU-MM01 (All Search)					
	Rank-1	Rank-5	Rank-10	Rank-20	mAP	mINP
0%	63.71	87.94	94.13	98.02	59.94	45.13
10%	61.70	85.75	92.72	97.17	58.21	43.80
30%	59.74	84.69	92.38	97.15	56.28	41.39
50%	49.22	76.97	85.31	92.67	45.34	30.04
70%	37.66	68.51	79.19	85.50	33.49	21.83

It is worth noting that BIMM-Net is primarily designed to address unsupervised cross-modal association within a given dataset, rather than cross-domain generalization. The reported results therefore help clarify the applicability and limitations of the proposed framework under cross-dataset deployment. Although cross-dataset performance is not the primary objective of this work, this analysis provides a clearer understanding of current unsupervised VI-ReID methods under domain shifts and may motivate future research toward domain-aware or adaptation-based extensions. Furthermore, while BIMM-Net targets visible-infrared person re-identification, its bi-level modulation and unsupervised cross-modal alignment schemes can be generalized to other visual tasks. Existing studies [49], [50] have shown the significance of general representation learning and cross-scenario generalization in visual tasks. Our core framework can be extended to more cross-modal and limited-label visual tasks in future work.

3) *Unpaired Settings*: To evaluate robustness under unpaired settings, we injected controlled noise into TPCC-based cross-modal matches by randomly swapping a given proportion of matched pairs on SYSU-MM01. The results are summarized in Tab. VI. As the noise ratio increases, performance gradually decreases, but BIMM-Net still maintains reasonable pseudo-label association under moderate

perturbations (10–30%), demonstrating that the bi-level mechanism effectively limits error propagation in unpaired or adversarial settings.

G. Hyperparameter Analysis

In the experiment, the parameter λ in Eq. 4 is randomly sampled from $[0, \lambda]$, so the set λ value determines the graying upper bound of GrayMix. A larger λ endows GrayMix with more diverse color variations and generates more low-chroma samples, which narrows the gap with IR images. Conversely, a smaller λ restricts the graying degree of GrayMix, making it more inclined to produce near-RGB images to enrich training resources. We tested the impact of different upper bounds of λ on the performance of SYSU-MM01 and RegDB datasets in Fig. 5 (a) and (b), respectively. As shown in Fig. 5 (a), when λ is set to 0.5 on the SYSU-MM01 dataset, the optimal performance is achieved. This is because the SYSU-MM01 dataset has a complex data distribution and significant cross-modal discrepancy, requiring a more conservative λ value to balance training difficulty and generalization performance. By contrast, the RegDB dataset, with its lower training difficulty, can accept richer inputs. The performance continuously improves as λ increases, reaching the optimal value when $\lambda = 1$. This indicates that as long as the model can keep up with the increasing training difficulty, a larger λ leads to a higher upper limit of generalization performance.

We also conduct experiments on the parameter β in Eq. 22, whose value determines the relative weight of mixture memory anchors and single-modality memory anchors for samples in cross-modal contrastive learning. When β increases, the optimization weight of single-modality clusters increases, pushing samples of the same identity to lean toward single-modality memory. When β decreases, it prompts samples of the same identity to lean toward mixture memory. As shown in Fig. 5 (c) and (d), the optimal performance is achieved when β is set to 0.5 on both SYSU-MM01 and RegDB datasets. This indicates that in cross-modal contrastive learning, the dominant role of mixture memory as an anchor helps the model learn modality-invariant features and promotes cross-modal semantic alignment.

Furthermore, we conduct an experimental analysis on the parameter μ in Eq. 29, whose value determines the weight of the HIA loss. Unlike the contrastive learning loss objective, the HIA loss does not directly optimize the model's sample recognition ability but promotes the semantic alignment of modality distributions in the mixture memory space. When μ

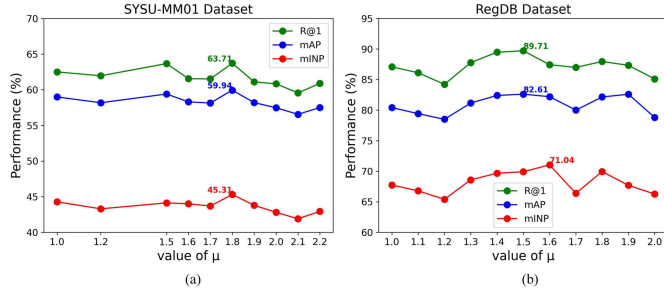


Fig. 6. Influence of μ values on SYSU-MM01 dataset under the All Search mode and RegDB dataset under the Visible to Infrared mode. Rank-1 accuracy(%), mAP (%) and mINP (%) are reported.

TABLE VII

SENSITIVITY ANALYSIS OF THE DBSCAN PARAMETER eps ON THE SYSU-MM01 DATASET (ALL SEARCH). THE BOLD FONT AND UNDERLINE FONT INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

eps	SYSU-MM01 (All Search)					
	Rank-1	Rank-5	Rank-10	Rank-20	mAP	mINP
0.45	61.58	85.53	92.68	97.29	57.33	41.78
0.50	61.64	86.06	93.03	97.32	57.85	43.15
0.55	<u>61.71</u>	85.22	92.10	96.83	<u>58.41</u>	43.83
0.60	63.71	87.94	94.13	98.02	59.94	45.13
0.65	56.09	83.55	91.92	97.28	54.20	40.24
0.70	50.69	77.90	87.79	95.31	48.75	34.94

TABLE VIII

SENSITIVITY ANALYSIS OF THE DBSCAN PARAMETER k ON THE SYSU-MM01 DATASET (ALL SEARCH). THE BOLD FONT AND UNDERLINE FONT INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

k	SYSU-MM01 (All Search)					
	Rank-1	Rank-5	Rank-10	Rank-20	mAP	mINP
4	63.71	87.94	94.13	98.02	59.94	45.13
6	61.48	86.06	93.26	98.01	57.91	43.25
8	59.88	84.93	92.51	97.47	57.07	42.87

increases, the model places greater emphasis on multi-modal distribution alignment. We tested the impact of different μ values on model performance for SYSU-MM01 and RegDB datasets in Fig. 6 (a) and (b), respectively. As shown in Fig. 6 (a), the model achieves optimal performance on SYSU-MM01 when μ is set to 1.8, a value higher than the optimal $\mu = 1.5$ for the RegDB dataset. The relatively large values of μ for both datasets indicate the importance of cross-modal distribution alignment in the USL-VI-ReID task. Moreover, the slightly higher optimal μ for SYSU-MM01 compared to RegDB suggests that datasets with higher training difficulty and more complex cross-modal discrepancies, such as SYSU-MM01, have a greater need for distribution alignment strategies.

In addition to the above parameters, we further analyze the sensitivity of the DBSCAN hyperparameters used in multiple stages of BIMM-Net, as they directly affect pseudo-label quality and cluster reliability. Specifically, we investigate the neighborhood radius eps and the minimum number of points k (i.e., minPts) on the SYSU-MM01 dataset, with results summarized in Tabs. VII and VIII.

As shown in Tabs. VII, increasing eps from a small value to a moderate range leads to consistent performance improvement, indicating that overly strict density constraints

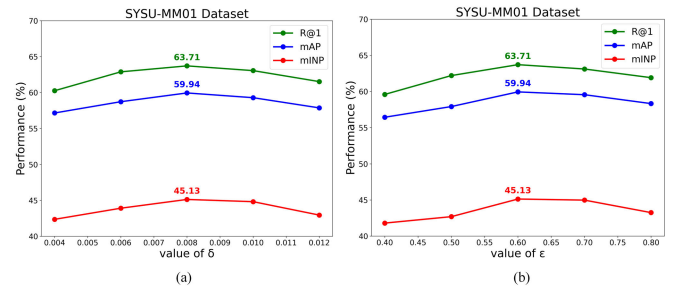


Fig. 7. Influence of δ and ϵ values on SYSU-MM01 dataset under the All Search mode. Rank-1 accuracy(%), mAP (%) and mINP (%) are reported.

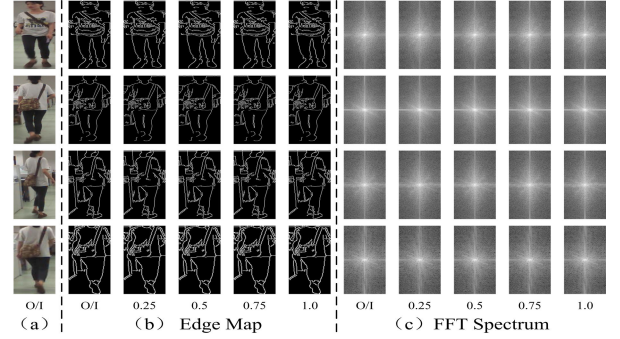


Fig. 8. (a) Visualization of the original RGB image; (b) The Canny edge map that highlights high-frequency structural boundaries; (c) The FFT spectrum that reveals the global frequency distribution.

tend to fragment clusters and suppress valid samples. The best performance is achieved at $eps = 0.60$, while nearby values (e.g., 0.50 and 0.55) also yield competitive results, suggesting a relatively stable operating range. When eps becomes too large, performance degrades due to the introduction of noisy neighborhood associations. Tab. VIII reports the influence of the parameter k . As k increases, overall performance gradually declines, implying that overly conservative density requirements reduce cluster coverage and limit effective pseudo-label propagation. The optimal performance is obtained at $k = 4$, which provides a favorable balance between cluster inclusiveness and density reliability.

Finally, we analyze the CNCC module, which uses dual thresholds δ and ϵ to select reliable cross-modal neighbors. As shown in Fig. 7, setting δ too small admits ambiguous pairs, while too large excludes valid ones; the optimal value is $\delta = 0.008$. Similarly, ϵ balances confidence and coverage, with $\epsilon = 0.6$ yielding the best results. These findings indicate that CNCC performs robustly within a reasonable parameter range, complementing the stability observed in the DBSCAN analysis.

H. Visualization Analysis

1) *Edge and Frequency Analysis*: To verify whether GrayMix leads to the loss of edge or high-frequency information, we additionally conducted two analyses: In Fig. 8, we present the results of our edge detection and Fast Fourier Transform (FFT) based analyses, where (a) shows the RGB original image (abbreviated as O/I), (b) displays the Canny edge maps obtained from both the O/I and the

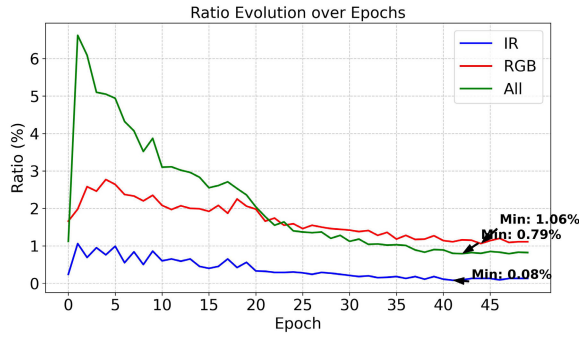


Fig. 9. Evolution of the ratio of boundary (low-confidence) samples identified by DBSCAN over training epochs on SYSU-MM01.

GrayMix-generated images under different values of λ , and (c) illustrates the corresponding FFT magnitude spectra. From the Canny results in (b), it is evident that the edge maps of the O/I and those of the GrayMix variants exhibit only small differences, indicating that GrayMix preserves structural or boundary information, as its operation affects only color composition rather than spatial gradients. Similarly, the FFT results in (c)—which include the spectra of the O/I and of the GrayMix images under multiple λ values—show that their frequency distributions remain highly consistent within the same image, with only minor variations.

This consistency further confirms that GrayMix preserves the intrinsic high-frequency components and avoids degrade texture-related information. Both experiments consistently demonstrate that GrayMix avoids damaging the discriminative high-frequency textures of the input images.

2) *Fragmentation Analysis*: To analyze potential fragmentation in low-density regions caused by DBSCAN, we visualize the evolution of boundary (low-confidence) samples during training, as shown in Fig. 9. Here, boundary samples are defined as the instances that are not density-reachable from any core points in the DBSCAN clustering process, characterized by low local density and ambiguous feature representation, thus being classified as low-confidence samples. At the initial stage, the boundary-sample ratio is relatively low because PGM-based intra-modal clustering produces compact clusters within each modality. However, due to the lack of explicit cross-modal alignment, these clusters remain poorly aligned across modalities.

After introducing GrayMix-based tri-modal matching and bi-level refinement, cross-modal consistency is explicitly optimized. This leads to a temporary increase and fluctuation in boundary samples during early epochs, reflecting the correction of cross-modal discrepancies rather than instability. As training proceeds, boundary samples are gradually absorbed into coherent clusters, and the boundary-sample ratio consistently decreases, converging to a very low level in later epochs.

3) *Feature Distribution*: Visualizing the feature distributions of visible and infrared images helps to intuitively understand the model's performance. Fig. 10 shows the t-SNE [51] feature distribution visualization of 20 randomly selected identities from the SYSU-MM01 dataset. Circles denote visible-light modality samples, crosses denote infrared-light modality samples, and different colors represent distinct

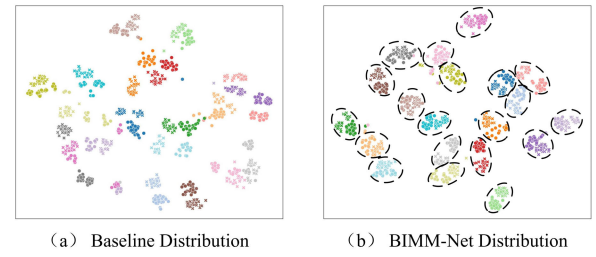


Fig. 10. Visually display the distribution of person features from the VIS and IR modalities on the SYSU-MM01 dataset in a two-dimensional feature space.

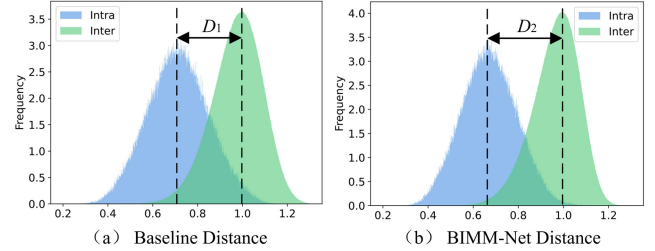


Fig. 11. Visually present the intra-class and inter-class similarity distributions of VIS and IR modality features on the SYSU-MM01 dataset.

identity categories. (a) is the feature distribution generated by the Baseline method: although some samples are effectively aggregated and cross-modal samples of the same identity show partial alignment, the intra-class compactness remains limited, and some samples fail to achieve ideal aggregation. (b) shows the distribution generated by BIMM-Net, which significantly improves intra-class compactness and further reduces the distance between cross-modal samples of the same identity compared to the Baseline. This demonstrates that BIMM-Net enhances both identity recognition capability and cross-modal semantic alignment ability.

4) *Feature Distance*: We further compared the intra-class and inter-class distribution distances between the Baseline and BIMM-Net. In classification tasks, an ideal feature distribution should exhibit high intra-class compactness and strong inter-class separability, which directly reflects the model's robustness in sample recognition. As shown in Fig. 11, it can be observed that the difference between inter-class and intra-class distances (D_2) of BIMM-Net is larger than that of the Baseline (D_1). This result not only mutually validates the t-SNE visualization in Fig. 10 but also confirms from the distribution perspective that BIMM-Net effectively enhances inter-class discriminability and intra-class compactness of features.

5) *Retrieval Result*: To further validate the efficacy of BIMM-Net, Fig. 12 presents a comparative analysis of retrieval rankings between the Baseline (B/L) and BIMM-Net under the all-search mode of the SYSU-MM01 dataset. In the experiment, infrared images serve as query images to retrieve corresponding visible images, where correct retrievals are marked with green bounding boxes and incorrect ones with red boxes. In (a), we show the comparison of retrieval results between BIMM-Net and the Baseline on samples with certain visual distinguishability, and it can be observed that BIMM-Net achieves a significantly higher retrieval



Fig. 12. Visualization of retrieval results obtained by the baseline (B/L) and our BIMM-Net on SYSU-MM01 under the All Search Mode. (a) Two groups of visually distinguishable normal samples. (b) Two groups of visually indistinguishable hard samples.

success rate than the Baseline. Among the samples successfully retrieved by BIMM-Net, obvious light-dark variations are visible. This is precisely because the GrayMix auxiliary modality is introduced during training, which simulates the depth transformation of color information, enabling the model to form a more robust understanding of color and light-dark relationships and thereby improving recognition performance. In (b), we further evaluate the retrieval performance of both methods on challenging hard samples. The upper group in (b) focuses on severe occlusion, where the query images are heavily blocked by objects such as chairs, trees, or vehicles. Even under such conditions, BIMM-Net retrieves more correct matches than the baseline, demonstrating stronger robustness to occlusion. The lower group in (b) examines extremely low-resolution and blurred query images, which contain very limited discriminative information. In this scenario, the baseline fails to retrieve any correct results, while BIMM-Net is still able to correctly identify one match, although its overall performance also degrades due to the lack of reliable visual cues.

V. CONCLUSION

This paper proposes a Bi-level Inter-modality Modulation Network, which effectively bridges the modality gap between visible and infrared images through the collaborative interaction of the single-modality clustering level and cross-modality clustering level. Extensive experimental results validate the effectiveness of BIMM-Net in mitigating modality discrepancies and enhancing the model's cross-modal recognition performance. However, when confronted with hard samples of low distinguishability, BIMM-Net still encounters difficulties in cross-modal retrieval. In future work, we will conduct further research on the issue of cross-modal hard sample pairs based on BIMM-Net.

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